Quantum Sensing and Metrology

Summary

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Contents

1 Mach–Zehnder interferometer **2**
1.1 Standard quantum limit . . . 3

2 Ramsey interferometer **3**

3 Classical parameter estimation **4**
3.1 Properties of estimators . . . 4
3.2 Properties of the Fisher information . . . 5
3.3 Maximum likelihood estimator . . . 5
3.4 Method of moments . . . 5
3.5 Bayesian estimation . . . 6

4 Gravitational wave detection **6**
4.1 Interferometric detection . . . 7

5 Quantum Mach–Zehnder interferometer **8**

6 Ramsey interferometry with spins **8**
6.1 Schwinger representation . . . 9

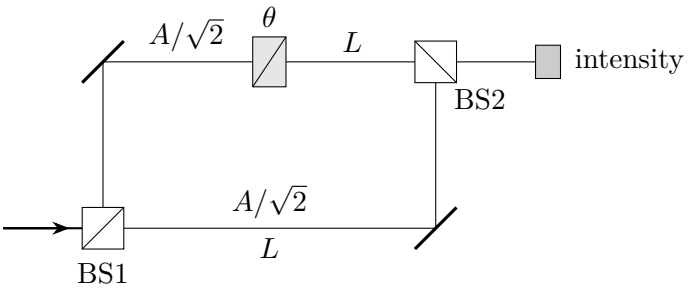
7 Squeezing **9**

8 SU(1,1) interferometer **10**

9 Quantum parameter estimation **10**
9.1 Properties of the QFI . . . 11

10 Entanglement and sensitivity **12**

1 Mach–Zehnder interferometer



We want to measure the phase θ. The output intensity is

$$I = \frac{1}{2}\left|\frac{A}{\sqrt{2}}e^{ikL} + \frac{A}{\sqrt{2}}e^{ikL+i\theta}\right|^2 = \frac{A^2}{2}\left(1+\cos\theta\right).$$

Sensitivity means: how well can we distinguish $I_{\theta+\delta\theta}$ from I_θ?

$$I_{\theta+\delta\theta} \simeq \frac{A^2}{2}(1+\cos\theta) - \frac{A^2}{2}\sin\theta\,\delta\theta.$$

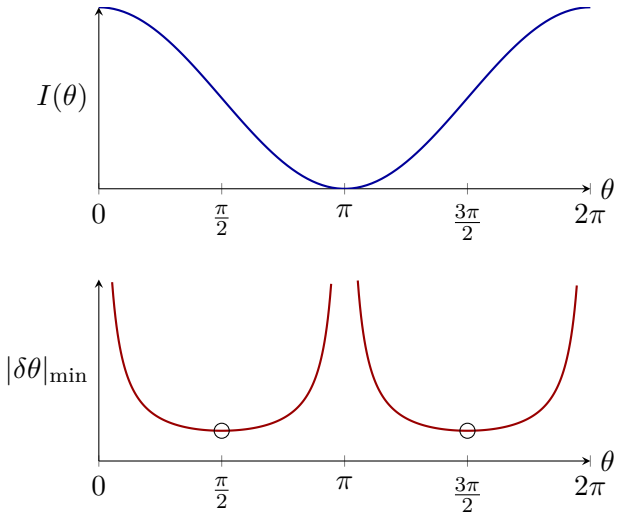
To be able to detect a signal we need $|I_{\theta+\delta\theta} - I_\theta| \geq \Delta I$, with ΔI the measurement resolution:

$$|\delta\theta| \geq \frac{\Delta I}{\frac{A^2}{2}\,|\sin\theta|}$$

so that

$$\min_\theta |\delta\theta| = \frac{\Delta I}{A^2/2} \qquad \text{at } \theta = \frac{\pi}{2} + n\pi.$$

To have good sensitivity we want a low ΔI, a large A, and $\theta \simeq \pi/2$, i.e. we work at the point of maximum slope of $I(\theta)$.



Circled: optimal working points, at maximum slope of $I(\theta)$.

We accomplish this with a bias θ_0:

$$I(\theta) = \frac{A^2}{2}(1+\cos(\theta-\theta_0)), \qquad \theta_0 \simeq \frac{\pi}{2} + \theta.$$

1.1 Standard quantum limit

Can ΔI be reduced arbitrarily? A laser is a coherent state $|\alpha\rangle$, with

$$\langle n\rangle = |\alpha|^2 = \bar{n}, \qquad \Delta n = \sqrt{\bar{n}}.$$

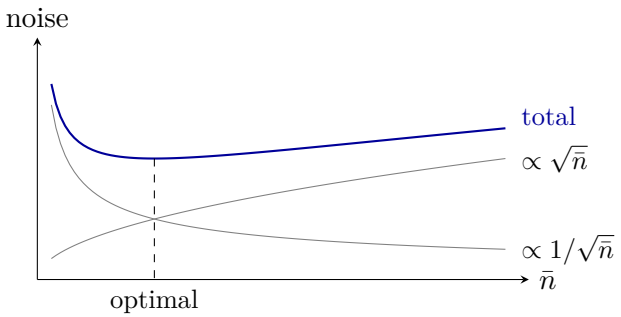
The minimum ΔI we can have, in order to at least distinguish a change in intensity from a random fluctuation, is one standard deviation, $\Delta I \sim \sqrt{\bar{n}}$. Hence the best possible sensitivity is

$$|\delta\theta| \sim \frac{\Delta I}{A^2/2} \sim \frac{\sqrt{\bar{n}}}{\bar{n}/2} \sim \frac{1}{\sqrt{\bar{n}}} \qquad \text{(standard quantum limit, SQL)}$$

If we try to increase the power, at some point we are limited by the momentum transfer of the photons to the mirrors (radiation pressure):

$$F = \frac{2P}{c}, \qquad \Delta F \propto \Delta N \propto \sqrt{\bar{n}}.$$

The two contributions, shot noise $\propto 1/\sqrt{\bar{n}}$ and radiation-pressure noise $\propto \sqrt{\bar{n}}$, give an optimal photon number:

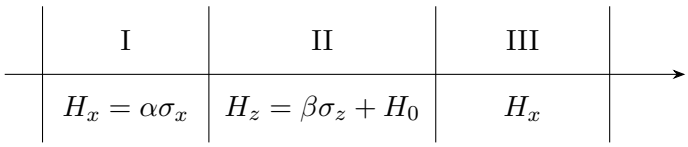


2 Ramsey interferometer

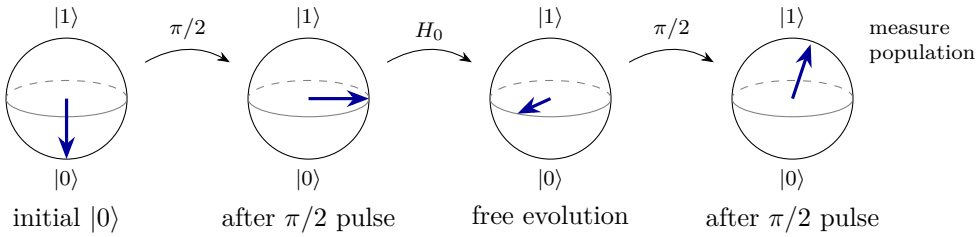
It is similar to the Mach–Zehnder interferometer. A two-level atom,

$$H_0 = E_0|0\rangle\langle 0| + E_1|1\rangle\langle 1|,$$

passes through three regions:



On the Bloch sphere, setting $\beta = 0$ and $t_3 = t_1 = \frac{\pi}{4\alpha}$, we have the following procedure:



Similarly to the MZI, we get

$$P(0) = \sin^2\!\left(\frac{(E_1 - E_0)\, t_2}{2}\right), \qquad \text{with } \alpha_{\text{opt}} = \frac{\hbar\pi}{4t}.$$

By measuring the oscillation frequency we can obtain $\Delta E = E_1 - E_0$.

3 Classical parameter estimation



The experiment is repeated m times to estimate a parameter θ, with i.i.d. outcomes. The probability of μ_i to occur is $P_i(\mu_i|\theta)$, and the LIKELIHOOD FUNCTION is

$$P(\boldsymbol{\mu}\,|\,\theta) = \prod_i P_i(\mu_i\,|\,\theta).$$

To estimate the value of θ from the measurement results $\boldsymbol{\mu}$ we construct an ESTIMATOR $\Theta = \Theta(\boldsymbol{\mu})$. Since it is a function of the random variable $\boldsymbol{\mu}$, it is itself a random variable, so it has a

- **mean value** $\langle\Theta\rangle = \bar{\Theta} = \sum_{\boldsymbol{\mu}} P(\boldsymbol{\mu}\,|\,\theta)\,\Theta(\boldsymbol{\mu})$;
- **variance** $(\Delta\Theta)^2 = \sum_{\boldsymbol{\mu}} P(\boldsymbol{\mu}\,|\,\theta)\,\big(\Theta(\boldsymbol{\mu}) - \langle\Theta\rangle\big)^2$.

The discrepancy between estimated and true value is given by the mean squared error

$$\text{MSE} = \sum_{\boldsymbol{\mu}} P(\boldsymbol{\mu}\,|\,\theta)\,\big(\Theta(\boldsymbol{\mu}) - \theta\big)^2 = \underbrace{\big(\langle\Theta\rangle - \theta\big)^2}_{\text{bias}^2} + (\Delta\Theta)^2.$$

This is the BIAS–VARIANCE TRADEOFF: reducing the bias increases the variance.

3.1 Properties of estimators

(i) **Unbiased**: if $\langle\Theta\rangle = \theta\ \forall\theta$ (and hence $\partial\bar{\Theta}/\partial\theta = 1$). If this holds only for a large number of measurements, the estimator is called *asymptotically unbiased*.

(ii) **Consistent**: if for $m \to \infty$

$$P\big(|\Theta(\boldsymbol{\mu}) - \theta| \geq \varepsilon\big) \to 0 \qquad \forall\varepsilon > 0.$$

A consistent estimator is asymptotically unbiased.

(iii) **Efficient**: if $\Theta(\boldsymbol{\mu})$ has the lowest possible variance, i.e. it saturates the CRAMÉR–RAO BOUND:

$$\boxed{(\Delta\Theta)^2 \geq \frac{\left|\partial\bar{\Theta}/\partial\theta\right|^2}{F(\theta)}}$$

where F is the FISHER INFORMATION

$$F(\theta) = \left\langle \left(\frac{\partial}{\partial\theta}\log P(\boldsymbol{\mu}\,|\,\theta)\right)^2 \right\rangle = \left\langle -\frac{\partial^2}{\partial\theta^2}\log P(\boldsymbol{\mu}\,|\,\theta) \right\rangle = \sum_{\boldsymbol{\mu}} \frac{1}{P(\boldsymbol{\mu}\,|\,\theta)} \left(\frac{\partial P(\boldsymbol{\mu}\,|\,\theta)}{\partial\theta}\right)^2.$$

The FI quantifies how much information the distribution $P(\boldsymbol{\mu}\,|\,\theta)$ carries about θ.

- For an unbiased estimator, $(\Delta\Theta)^2_{\mathrm{CRB}} = 1/F$.
- For m independent measurements, $F_{\mathrm{tot}} = \sum_i F_i$.
- The CRB can be saturated by an estimator if and only if

$$\frac{\partial}{\partial\theta}\log P(\boldsymbol{\mu}\,|\,\theta) = \lambda(\Theta(\boldsymbol{\mu}) - \langle\Theta\rangle) \quad \forall\boldsymbol{\mu}, \qquad \lambda = \frac{F(\theta)}{\partial\langle\Theta\rangle/\partial\theta}.$$

3.2 Properties of the Fisher information

(a) **Convexity**: for $P(\boldsymbol{\mu}\,|\,\theta) = \sum_k \lambda_k P_k(\boldsymbol{\mu}|\theta)$ with $\lambda_k \geq 0$, $\sum_k \lambda_k = 1$,

$$F(\theta) \leq \sum_k \lambda_k F_k(\theta).$$

(b) **Additivity**: $F(\theta) = \sum_{i=1}^m F_i(\theta)$ for m measurements.

Since computing F requires full knowledge of $P(\boldsymbol{\mu}\,|\,\theta)$, we can give a lower bound. Consider an observable $M(\mu)$ with $\bar{M}_\theta = \sum_\mu P(\mu|\theta)M(\mu)$; then

$$\boxed{F(\theta) \geq \frac{1}{(\Delta M)^2}\left(\frac{d\bar{M}}{d\theta}\right)^2}$$

3.3 Maximum likelihood estimator

$$\boxed{\Theta_{\mathrm{MLE}}(\boldsymbol{\mu}) = \arg\max_\phi P(\boldsymbol{\mu}\,|\,\phi)}$$

Properties of the MLE:

(i) **Consistency**: $\Theta_{\mathrm{MLE}}(\boldsymbol{\mu}) \xrightarrow[m\to\infty]{P} \theta$.

(ii) **Efficiency**: the distribution of values $\Theta_{\mathrm{MLE}}(\boldsymbol{\mu})$ tends to a Gaussian with variance $1/F$ by the central limit theorem: $P(\Theta_{\mathrm{MLE}}|\theta) \to \mathcal{G}$.

Hence the MLE is asymptotically unbiased, consistent and efficient. It is best when we have a lot of data, i.e. when we know $P(\boldsymbol{\mu}\,|\,\theta)$ well.

3.4 Method of moments

Consider an observable M with mean $\langle M\rangle = f(\theta)$ and variance ΔM^2, and the sample average

$$M_m = \frac{1}{m}\sum_{i=1}^m \mu_i.$$

By the CLT, $M_m \xrightarrow{m\to\infty}$ Gaussian. The estimator is

$$\boxed{\Theta_{\mathrm{MOM}} \coloneqq f^{-1}(M_m)}$$

It is asymptotically unbiased and asymptotically efficient.

3.5 Bayesian estimation

This is fundamentally different from the other approaches. We start with a PRIOR that gives the current knowledge about the true value of a parameter. After a measurement, we update it by computing a POSTERIOR probability through Bayes' theorem:

$$\underbrace{P(\phi \,|\, \boldsymbol{\mu})}_{\text{posterior}} = \frac{P(\boldsymbol{\mu} \,|\, \phi)\overbrace{P(\phi)}^{\text{prior}}}{\underbrace{P(\boldsymbol{\mu})}_{\text{normalization}}}.$$

$P(\phi)$ represents the probability that $\phi = \theta$ based on current knowledge. There are different types of estimators:

- **Posterior mean**: $\bar{\Theta}_{\text{BE}} = \int_a^b d\phi\, P(\phi|\boldsymbol{\mu})\, \phi$;
- **Absolute maximum**: $\Theta_{\text{BE}}^{\max} = \arg\max_{\phi\in[a,b]} P(\phi|\boldsymbol{\mu})$ (for a flat prior it is the same as the MLE).

From one of these point estimates we can compute the POSTERIOR VARIANCE

$$(\Delta\phi)^2 \coloneqq \int_a^b d\phi\, P(\phi|\boldsymbol{\mu})(\phi - \Theta_{\text{BE}}(\boldsymbol{\mu}))^2,$$

which specifies a confidence interval around Θ_{BE} for a specific measurement sequence $\boldsymbol{\mu}$. It is different from the estimator variance, which measures how much the output of the estimator fluctuates when repeating the measurements. From $P(\phi|\boldsymbol{\mu})$ we can also find confidence intervals Ω:

$$P(\phi \in \Omega \,|\, \boldsymbol{\mu}) = \int_\Omega d\phi\, P(\phi|\boldsymbol{\mu}).$$

The Bayesian method is asymptotically consistent and efficient.

We can get a bound that applies to the posterior variance, called the GOSH BOUND:

$$(\Delta\phi)^2 \geq \frac{1}{G(\boldsymbol{\mu})}, \qquad G(\boldsymbol{\mu}) = \int_a^b d\phi\, \frac{1}{P(\phi|\boldsymbol{\mu})}\left(\frac{dP(\phi|\boldsymbol{\mu})}{d\phi}\right)^2$$

It is similar to the CRB, but:

- $F(\theta)$ measures how sensitive your measurement is to changes in the parameter; it is a property of the system and does not depend on $\boldsymbol{\mu}$;
- $G(\boldsymbol{\mu})$ measures how much information has been extracted by a specific measurement $\boldsymbol{\mu}$. It is not additive.

For $m \to \infty$, the LAPLACE–BERNSTEIN–VON MISES THEOREM states that $P(\phi|\boldsymbol{\mu})$ becomes a Gaussian centred at the true θ with variance $1/F(\theta)$, so the Gosh bound $\xrightarrow{m\to\infty}$ CRB.

4 Gravitational wave detection

In general relativity, mass and energy curve space-time:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \frac{8\pi G}{c^4}T_{\mu\nu}.$$

Distance is expressed as

$$ds^2 = \sum_{\mu\nu} g_{\mu\nu}\, dx^\mu dx^\nu,$$

which in empty space (Minkowski metric $\eta_{\mu\nu}$) reads

$$ds^2 = -c^2dt^2 + dx^2 + dy^2 + dz^2.$$

We can expand to linear order, $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ with $|h_{\mu\nu}| \ll 1$, outside the source ($T_{\mu\nu} = 0$), and choose a coordinate system such that $h = \operatorname{tr} h_{\mu\nu} = 0$:

$$-\frac{1}{c^2}\frac{\partial^2}{\partial t^2}h_{\mu\nu} + \nabla^2 h_{\mu\nu} = 0.$$

The solutions are plane waves called GRAVITATIONAL WAVES,

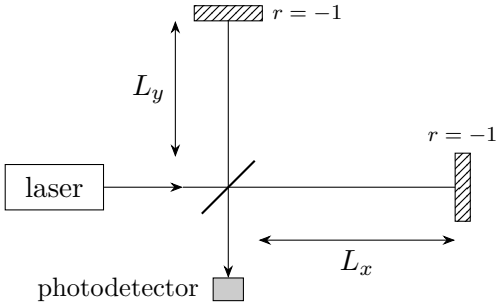
$$h_{\mu\nu} = e_{\mu\nu}\, e^{i(\omega t - \mathbf{k}\cdot\mathbf{x})},$$

with $e_{\mu\nu}$ the polarization tensor. There are two possible polarizations h_+, $h_\times$, transverse to the propagation:

$$h_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_\times & 0 \\ 0 & h_\times & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

4.1 Interferometric detection

How to measure them? With an interferometer:



$$P \propto E_{\text{out}}^2 = E_0^2 \sin^2\big(k_L(L_x - L_y)\big).$$

Consider a (+)-polarized wave coming from the z direction, $h_+(t) = h_0 \cos(\omega_{\text{GW}} t)$:

$$\begin{aligned} ds^2 &= -c^2dt^2 + \big(1 + h_+(t)\big)dx^2 + \big(1 - h_+(t)\big)dy^2 + dz^2 \\ \Rightarrow \quad L_x &\to L_x\big(1 + \tfrac{1}{2}h_+(t)\big), \qquad L_y \to L_y\big(1 - \tfrac{1}{2}h_+(t)\big) \\ \Rightarrow \quad P &\propto E_{\text{out}}^2 = E_0^2 \sin^2\big(k_L L\, h_+(t) + \phi_0\big). \end{aligned}$$

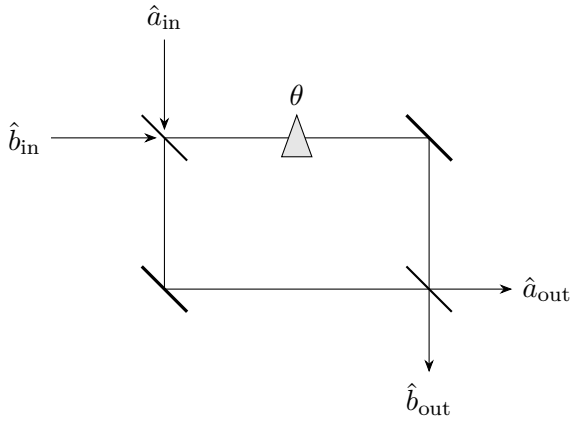
The intensity oscillates when a GW passes. To increase the sensitivity we increase L with a Fabry–Pérot cavity of finesse $\mathcal{F}$:

$$\tilde{L}_x = \frac{2}{\pi}\mathcal{F}L_x, \qquad \mathcal{F} = \mathcal{O}(100) \text{ at LIGO}.$$

We then have the usual bound and the SQL:

$$\delta\theta \geq \frac{\Delta I}{\frac{A^2}{2}\,|\sin(\theta - \theta_0)|}, \qquad \delta\theta \sim \frac{1}{\sqrt{\bar{n}}}.$$

5 Quantum Mach–Zehnder interferometer



We use the Heisenberg picture: operators transform, not the quantum states.

$$\begin{pmatrix}\hat{a}_{\rm out}\\ \hat{b}_{\rm out}\end{pmatrix} = \underbrace{\frac{1}{\sqrt{2}}\begin{pmatrix}1 & i\\ i & 1\end{pmatrix}}_{\rm BS2}\begin{pmatrix}e^{i\theta} & 0\\ 0 & 1\end{pmatrix}\underbrace{\frac{1}{\sqrt{2}}\begin{pmatrix}1 & -i\\ -i & 1\end{pmatrix}}_{\rm BS1}\begin{pmatrix}\hat{a}_{\rm in}\\ \hat{b}_{\rm in}\end{pmatrix}$$
$$= e^{i\theta/2}\begin{pmatrix}\cos\frac{\theta}{2} & \sin\frac{\theta}{2}\\ -\sin\frac{\theta}{2} & \cos\frac{\theta}{2}\end{pmatrix}\begin{pmatrix}\hat{a}_{\rm in}\\ \hat{b}_{\rm in}\end{pmatrix}.$$

We can measure

$$\langle N_{a,\rm out}\rangle = \langle \hat{a}_{\rm in}^\dagger \hat{a}_{\rm in}\rangle \cos^2\frac{\theta}{2} + \langle \hat{b}_{\rm in}^\dagger \hat{b}_{\rm in}\rangle \sin^2\frac{\theta}{2},$$

and similarly for $\langle N_{b,\rm out}\rangle$. If $a_{\rm in}$ has an average of $\bar{N}$ photons and $b_{\rm in}$ is vacuum,

$$\langle N_{a,\rm out}\rangle = \frac{\bar{N}}{2}(1+\cos\theta), \qquad \langle N_{b,\rm out}\rangle = \frac{\bar{N}}{2}(1-\cos\theta),$$

and with $M = N_{a,\rm out} - N_{b,\rm out}$

$$\langle M\rangle = \bar{N}\cos\theta \quad \Rightarrow \quad (\Delta\theta)^2 = \frac{(\Delta M)^2}{|\partial\langle M\rangle/\partial\theta|^2}.$$

6 Ramsey interferometry with spins

Consider a collection of N spin-$\frac{1}{2}$ particles with collective spin

$$\mathbf{J} = \sum_{i=1}^{N}\frac{\boldsymbol{\sigma}^{(i)}}{2}.$$

A quantum state of the collective spin can be written as

$$|\psi\rangle = \sum_J\sum_m c_{Jm}|J,m\rangle, \qquad J = \tfrac{N}{2}, \tfrac{N}{2}-1, \ldots, \quad m = -J, \ldots, J.$$

The $|J,m\rangle$ are the DICKE STATES and they form a basis:

$$\mathbf{J}^2|J,m\rangle = J(J+1)|J,m\rangle, \qquad J_z|J,m\rangle = m|J,m\rangle.$$

The Ramsey interferometer consists of

$$|\psi_{\rm out}\rangle = e^{i\frac{\pi}{2}J_x}\, e^{-i\theta J_z}\, e^{-i\frac{\pi}{2}J_x}|\psi_{\rm in}\rangle.$$

In the Heisenberg picture this is

$$\begin{pmatrix} J_x \\ J_y \\ J_z \end{pmatrix}_{\text{out}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} J_x \\ J_y \\ J_z \end{pmatrix}_{\text{in}},$$

$$\langle J_z \rangle_{\text{out}} = \cos\theta \langle J_z \rangle_{\text{in}} - \sin\theta \langle J_x \rangle_{\text{in}}.$$

If we use the method of moments with $M = J_{z,\text{out}}$, then $(\Delta\theta)^2 = \dfrac{(\Delta M)^2}{(\partial\langle M\rangle/\partial\theta)^2}$ is easy to find.

6.1 Schwinger representation

To unify the Ramsey interferometer and the MZI we use the Schwinger representation:

$$J_x = \frac{1}{2}\left(a^\dagger b + b^\dagger a\right), \qquad J_y = \frac{1}{2i}\left(a^\dagger b - b^\dagger a\right), \qquad J_z = \frac{1}{2}\left(a^\dagger a - b^\dagger b\right),$$

where from $[a, a^\dagger] = [b, b^\dagger] = 1$ we get the SU(2) relations

$$[J_k, J_l] = i\varepsilon_{klm} J_m.$$

The operations in the MZI can now be seen as rotations in spin space:

$$\text{BS1: } U = e^{-i\frac{\pi}{2}J_x}, \qquad \text{BS2: } U = e^{i\frac{\pi}{2}J_x}, \qquad \text{PS: } U = e^{-i\theta J_z}.$$

For example, an interferometer with N photons in a and zero in b:

$$|N\rangle_a |0\rangle_b \;\rightarrow\; |J = \tfrac{N}{2},\, m = \tfrac{N}{2}\rangle.$$

Measuring $\langle J_z \rangle = \frac{1}{2}\left(\langle N_a \rangle - \langle N_b \rangle\right)$ is equivalent to measuring $\langle M \rangle$.

From $|\alpha\rangle|0\rangle$, with $\langle N_a \rangle = |\alpha|^2$, we get

$$(\Delta\theta)^2 = \frac{1}{\langle N_a \rangle |\sin\theta|^2},$$

the same as the classical description. With squeezing, in the limit of large N, we can beat the SQL $(\Delta\theta)^2 \sim 1/N$ and get

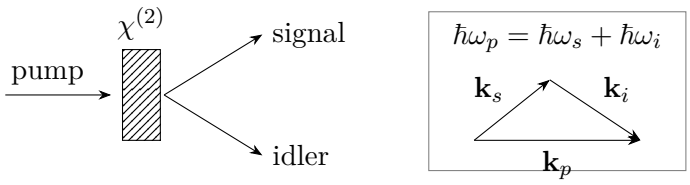
$$(\Delta\theta)^2 \sim \frac{1}{N^{3/2}}$$

with the same resources, i.e. the same total number of photons N.

7 Squeezing

To generate squeezed light, we send it through a nonlinear crystal:

$$H = i\hbar g\left(e^{-i\phi}\,\hat{a}\hat{b} - e^{i\phi}\,\hat{a}^\dagger\hat{b}^\dagger\right).$$



Two cases:

- **Degenerate**, $\omega_s = \omega_i$:

$$H = \hbar g' \left(\hat{a}^{\dagger 2} + \hat{a}^2\right) \;\rightsquigarrow\; S_1(\xi) = e^{-\frac{1}{2}\left(\xi^* \hat{a}^2 - \xi \hat{a}^{\dagger 2}\right)} \qquad \text{(single-mode squeezing)}$$

where $\xi \sim gt$, $\xi = re^{i\theta}$. The state $|\xi\rangle = S_1(\xi)|0\rangle$ has only even Fock states populated.

$$\hat{a}(t) = \hat{a}_0 \cosh r - i\hat{a}_0^\dagger e^{i\theta} \sinh r,$$
$$\hat{a}^\dagger(t) = \hat{a}_0^\dagger \cosh r + i\hat{a}_0 e^{-i\theta} \sinh r,$$

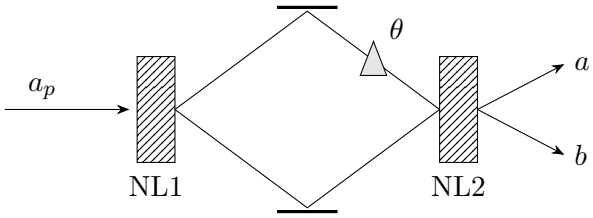
$$\Rightarrow \quad (\Delta X)^2 = \frac{1}{2}e^{-2r}, \qquad (\Delta P)^2 = \frac{1}{2}e^{2r}.$$

- **Non-degenerate**, $\omega_s \neq \omega_i$:

$$H = \hbar g' \left(\hat{a}\hat{b} + \hat{a}^\dagger \hat{b}^\dagger\right) \;\rightsquigarrow\; S_2(\xi) = e^{\left(\xi^* \hat{a}\hat{b} - \xi \hat{a}^\dagger \hat{b}^\dagger\right)} \qquad \text{(two-mode squeezing)}$$

$$S_2(\xi)|0\rangle = c_{00}|0,0\rangle + c_{11}|1,1\rangle + \ldots$$

8 SU(1,1) interferometer



We introduce the operators

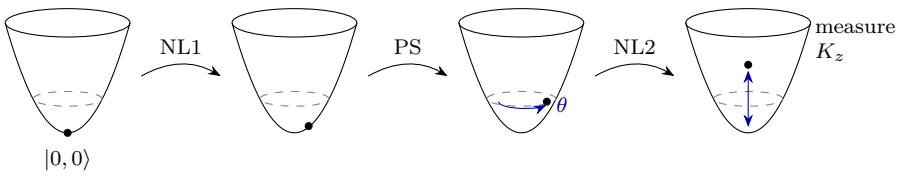
$$K_x = \frac{1}{2}\left(a^\dagger b^\dagger + ab\right), \qquad K_y = \frac{1}{2i}\left(a^\dagger b^\dagger - ab\right), \qquad K_z = \frac{1}{2}\left(a^\dagger a + b^\dagger b + 1\right).$$

These satisfy the SU(1,1) algebra. The conserved quantity is

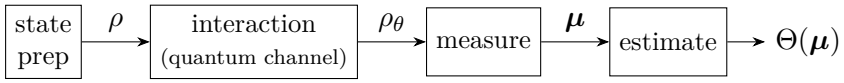
$$K^2 = K_z^2 - K_x^2 - K_y^2,$$

related to the photon-number difference (invariance). Then, instead of the Bloch sphere, states live on the surface of a hyperboloid. The total photon number is not constant.

$$|\psi_{\text{out}}\rangle = e^{irK_x}\, e^{-i\theta K_z}\, e^{-irK_x} |\psi_{\text{in}}\rangle.$$



9 Quantum parameter estimation



In quantum mechanics, using the Born rule,

$$P(\mu \,|\, \theta) = \mathrm{Tr}\left[E_\mu \rho_\theta\right],$$

where $\{E_\mu\}$ is a POVM:

$$\underbrace{\sum_\mu E_\mu = \mathbb{1}}_{\text{normalization}}, \qquad \underbrace{E_\mu \geq 0}_{\Rightarrow P(\mu|\theta)\geq 0} \quad \text{(positive semidefinite)}.$$

The classical Fisher information is a function of the probability distribution $P(\mu|\theta)$, which now depends on the choice of $\{E_\mu\}$:

$$F(\theta) = \sum_\mu \frac{1}{P(\mu|\theta)} \left(\frac{\partial P(\mu|\theta)}{\partial \theta}\right)^2 = F(\theta, \{E_\mu\}).$$

We define the QUANTUM FISHER INFORMATION, the absolute maximum of information we can extract:

$$\boxed{F_Q(\theta) = \max_{\{E_\mu\}} F(\theta, \{E_\mu\}) \qquad \rightsquigarrow \qquad (\Delta\theta)^2_{\mathrm{QCRB}} \geq \frac{1}{\nu\, F_Q(\theta)}}$$

with ν the number of repetitions. In general,

$$(\Delta\theta)^2_{\mathrm{exp}} \geq (\Delta\theta)^2_{\mathrm{CRB}} \geq (\Delta\theta)^2_{\mathrm{QCRB}}.$$

For a **pure state** undergoing the **unitary evolution** $U_\theta = e^{-i\theta H}$, where H is the generator of the evolution, we find

$$\boxed{F_Q(\theta) = 4\,\mathrm{Var}[H]}$$

In general, for a mixed state the QFI is lower, and for $\rho = \sum_k \lambda_k |k\rangle\langle k|$

$$\boxed{F_Q = 2 \sum_{\substack{k,l \\ \lambda_k+\lambda_l\neq 0}} \frac{(\lambda_k - \lambda_l)^2}{\lambda_k + \lambda_l} \,|\langle k|H|l\rangle|^2}$$

9.1 Properties of the QFI

(i) **Convexity**: if $\rho = \sum_k p_k \rho^{(k)}$ with $p_k \geq 0$, $\sum_k p_k = 1$,

$$F_Q[\rho, H] \leq \sum_k p_k F_Q[\rho^{(k)}, H].$$

(ii) **Additivity**: for m independent subsystems, $\rho = \rho^{(1)} \otimes \cdots \otimes \rho^{(m)}$,

$$F_Q[\rho, H] = m\, F_Q[\rho^{(1)}, H].$$

For example, for the vacuum state $\mathrm{Var}[X] = \mathrm{Var}[P] = \frac{1}{4}$. By squeezing one quadrature we get increased sensitivity in the other: for a displacement along x,

$$F_Q = 4\,\mathrm{Var}[P],$$

which can be increased by squeezing.

10 Entanglement and sensitivity

Consider N identical subsystems, and assume a perturbation that acts locally on each subsystem:

$$H = \sum_i h_i \quad \rightsquigarrow \quad U = \prod_i e^{-i\theta h_i}.$$

The max/min eigenvectors and eigenvalues of h_i are $|h_{\max}\rangle$, $h_{\max}$ and $|h_{\min}\rangle$, $h_{\min}$. We consider three cases.

Separable states.

$$\rho_{\text{sep}} = \sum_k p_k |\psi_{\text{sep}}^{(k)}\rangle\langle\psi_{\text{sep}}^{(k)}|, \qquad |\psi_{\text{sep}}^{(k)}\rangle = |\psi_1^{(k)}\rangle \cdots |\psi_N^{(k)}\rangle.$$

We find that

$$\boxed{F_Q[\rho_{\text{sep}}, H] \leq N\,(h_{\max} - h_{\min})^2}$$

This is called the STANDARD QUANTUM LIMIT, and gives a scaling

$$(\Delta\theta)^2_{\text{QCRB}} \sim \frac{1}{N}.$$

In this case the optimal state is

$$|\psi_{\text{opt}}\rangle = \frac{1}{\sqrt{2}}\big(|h_{\max}\rangle + |h_{\min}\rangle\big)$$

(on each subsystem).

Entangled states. Consider $\rho = \sum_k p_k |\psi^{(k)}\rangle\langle\psi^{(k)}|$ not separable. We find that, $\forall\rho$,

$$\boxed{F_Q[\rho, H] \leq N^2\,(h_{\max} - h_{\min})^2}$$

This is called the HEISENBERG LIMIT and gives

$$(\Delta\theta)^2_{\text{QCRB}} \sim \frac{1}{N^2}.$$

Mixture of separable/entangled states. If k is the size of the largest entangled group, $s = \lfloor N/k \rfloor$ and $r = N - sk$,

$$\boxed{F_Q[\rho, H] \leq \left(sk^2 + r^2\right)(h_{\max} - h_{\min})^2}$$

